

Day 01: Introduction to Python

1. What is Python?

Python is a high-level, interpreted, general-purpose programming language. It is popular because of its simple syntax and easy readability.

Python is used by beginners as well as professionals for building applications, data analysis, artificial intelligence, machine learning, automation, and web development.

2. History of Python

- Python was created by Guido van Rossum.
- Python was first released in 1991.
- The name "Python" was inspired by the comedy show "Monty Python's Flying Circus".

3. Why Learn Python?

Python is one of the most popular programming languages because:

- It is easy to learn and understand.
- It has simple and readable syntax.
- It has a large community support.
- It provides many built-in libraries.
- It is widely used in AI and Machine Learning.

4. Features of Python

Simple Syntax

Python code is easy to read and write.

Example:

```
print("Hello Python")
```

Interpreted Language

Python executes code line by line using an interpreter.

Platform Independent

Python programs can run on different operating systems like Windows, Linux, and macOS.

Open Source

Python is free to use and modify.

Object-Oriented Programming

Python supports classes and objects.

5. Applications of Python

Python is used in many fields:

Artificial Intelligence (AI)

Used to create intelligent systems.

Machine Learning

Used for building predictive models.

Data Science

Used for data analysis and visualization.

Web Development

Used to create websites and web applications.

Automation

Used to automate repetitive tasks.

6. Installing Python

To start programming in Python:

1. Download Python from the official website.
2. Install Python on your system.
3. Check installation using:

python --version

7. Python Development Tools

VS Code

A popular code editor for writing Python programs.

Jupyter Notebook

Used for interactive coding and data science projects.

Google Colab

A cloud-based platform for running Python programs.

8. First Python Program

Example:

```
print("Hello, World!")
```

Output:

Hello, World!

The print() function is used to display output on the screen.

9. Python Comments

Comments are used to explain code. Python ignores comments while execution.

Single-line comment:

```
# This is a comment  
print("Python")
```

10. Python Syntax Basics

Python uses indentation to define blocks of code.

Example:

```
if 10 > 5:  
    print("10 is greater")
```

Proper indentation is important in Python.

Summary

In Day 01, we learned:

- Introduction to Python
- History of Python
- Features and applications
- Python installation
- Development tools
- First Python program
- Comments and syntax basics

